Language Model Quiz

Questions, hints and explained answers

A readable study copy derived from the original native NotebookLM Quiz interface. The questions, answer choices, hints and explanations retain the observed source content. This document is a static PDF.

Quiz

Choose one answer for each question. Hints and solutions follow in separate sections.

1. A user inputs a explicit definition into a prompt (such as "In this story, X means Y"), and the model uses this definition in its immediate response. Does this interaction indicate that the model's parameters have changed?

A. Yes, because autoregressive language models continuously optimize their pretrained weights after every processed message.

B. Yes, any new detail successfully reflected in a generated response confirms that the model's parameters were updated during inference.

C. No, because parameter updates can only occur when Retrieval-Augmented Generation (RAG) is explicitly enabled.

D. No, inserting information into a prompt changes only the context available to that specific inference step, leaving model parameters fixed.

2. Why is it incorrect to assume that a word like "generative" will always correspond to exactly one token across language models?

A. Token segmentation depends on the specific tokenizer used and exact string details, including capitalization and preceding spaces.

B. Tokenization algorithms split text strictly by individual characters, so multi-letter words always yield multiple tokens.

C. Language models automatically re-segment words into tokens at every decoder layer based on positional encodings.

D. Vocabulary IDs are assigned alphabetically, forcing words with more than five letters to split into two tokens.

3. What is the key functional difference between a numerical token ID and its initial embedding vector?

A. A token ID measures the semantic closeness of words, whereas the embedding vector merely records positional sequence order.

B. A token ID is computed by the softmax layer, while the embedding is produced by causal attention masks.

C. A token ID changes dynamically depending on context, whereas the initial embedding remains identical across different sentences.

D. A token ID is an arbitrary lookup index in a vocabulary, whereas an initial embedding is a learned numerical vector retrieved by that ID.

4. A language model generates a candidate next token with a calculated probability of 80%. How should this percentage be correctly interpreted?

A. It indicates that 80% of the decoder blocks in the network voted in favor of choosing this specific token.

B. It represents the normalized statistical continuation likelihood over the model's vocabulary given the context, not a measure of factual truth.

C. It confirms an 80% statistical probability that the outputted statement is factually accurate.

D. It guarantees that 80% of pretraining documents contained this exact full sentence.

5. If the temperature parameter *T* is increased while keeping the context and output logits fixed, how does the resulting output probability distribution change?

A. The distribution concentrates heavily on the top logit, reducing the probability mass assigned to secondary alternatives.

B. The total sum of all probabilities over the vocabulary expands beyond 1.0.

C. The distribution becomes more uniform across alternatives, while keeping the original ranking order of the logits unchanged.

D. The logit scores are recalculated, changing which token holds the highest rank in the vocabulary.

6. When a system uses Retrieval-Augmented Generation (RAG) to incorporate an external document into a query, what is happening technically within the system?

A. The tokenizer vocabulary is updated to permanently add new vocabulary IDs found in the document.

B. The system uses the retrieved passage to execute a fast fine-tuning step that updates the model's weights.

C. The model's causal attention mask is disabled to allow direct insertion of missing factual information into output logits.

D. Selected external text passages are inserted into the prompt context for the fixed model to use during inference.

Hints

1. Distinguish between temporary information passed in an input context during inference and permanent structural updates to model weights.

2. Consider how text processing tools split text strings into numerical indices before vector representations are constructed.

3. Recall how initial lookup tables map discrete index values into multi-dimensional numerical vector spaces.

4. Reflect on what the output softmax calculation mathematically measures when predicting sequence continuations.

5. Think about how dividing logits by a larger temperature value affects the relative differences between exponentials in the softmax formula.

6. Focus on the structural distinction between altering the available input context and modifying fixed model parameters.

Solutions and explanations

Question 1

Correct answer: D

A. Yes, because autoregressive language models continuously optimize their pretrained weights after every processed message.

Pretraining and fine-tuning are separate training stages; standard inference processes user inputs without altering model weights.

B. Yes, any new detail successfully reflected in a generated response confirms that the model's parameters were updated during inference.

This confuses temporary context processing during inference with parameter optimization during training or fine-tuning.

C. No, because parameter updates can only occur when Retrieval-Augmented Generation (RAG) is explicitly enabled.

Retrieval-augmented generation supplies external documents into the context window, which also leaves model parameters completely unchanged.

D. No, inserting information into a prompt changes only the context available to that specific inference step, leaving model parameters fixed.

Inference runs computations using pre-existing parameters, so providing new information in a user prompt only alters the active context window rather than updating model weights.

Question 2

Correct answer: A

A. Token segmentation depends on the specific tokenizer used and exact string details, including capitalization and preceding spaces.

Tokens, words, and characters are distinct units; tokenization rules vary by tokenizer implementation and depend strictly on input string formatting.

B. Tokenization algorithms split text strictly by individual characters, so multi-letter words always yield multiple tokens.

Tokenizers often map whole words or subword chunks to single tokens rather than breaking every word down into individual characters.

C. Language models automatically re-segment words into tokens at every decoder layer based on positional encodings.

Tokenization is a discrete preprocessing step performed prior to initial vector embedding lookups, not an operation inside decoder layers.

D. Vocabulary IDs are assigned alphabetically, forcing words with more than five letters to split into two tokens.

Vocabulary identification numbers are index lookup values rather than character-length measurements or alphabetical rankings.

Question 3

Correct answer: D

A. A token ID measures the semantic closeness of words, whereas the embedding vector merely records positional sequence order.

Nearby token IDs do not imply similar meanings, and positional information is provided by separate positional encodings or mechanisms like RoPE.

B. A token ID is computed by the softmax layer, while the embedding is produced by causal attention masks.

Token IDs exist prior to model inference, while softmax operates at the output stage to convert logits into probabilities.

C. A token ID changes dynamically depending on context, whereas the initial embedding remains identical across different sentences.

The initial embedding retrieved by a token ID is fixed for that token ID; it becomes a contextual representation only after passing through decoder layers.

D. A token ID is an arbitrary lookup index in a vocabulary, whereas an initial embedding is a learned numerical vector retrieved by that ID.

Token IDs represent discrete indices without built-in semantic distance, while embeddings provide continuous vector representations that enter the model's neural transformations.

Question 4

Correct answer: B

A. It indicates that 80% of the decoder blocks in the network voted in favor of choosing this specific token.

The percentage is calculated via a softmax function over vocabulary logits at the final output layer, not by counting votes across network layers.

B. It represents the normalized statistical continuation likelihood over the model's vocabulary given the context, not a measure of factual truth.

Softmax probabilities reflect next-token likelihood distributions based on context and parameters; high probability does not guarantee factual accuracy.

C. It confirms an 80% statistical probability that the outputted statement is factually accurate.

High continuation probability predicts expected word sequences according to learned patterns, not factual correctness.

D. It guarantees that 80% of pretraining documents contained this exact full sentence.

Probabilities are calculated dynamically step-by-step over the vocabulary based on current logits, not calculated as direct training set frequency counts.

Question 5

Correct answer: C

A. The distribution concentrates heavily on the top logit, reducing the probability mass assigned to secondary alternatives.

Concentrating probability onto the top alternative occurs when lowering temperature, whereas raising temperature flattens the distribution.

B. The total sum of all probabilities over the vocabulary expands beyond 1.0.

Softmax normalizes logit exponent values so that the sum across the entire vocabulary always equals 1.0 regardless of temperature.

C. The distribution becomes more uniform across alternatives, while keeping the original ranking order of the logits unchanged.

Higher temperature scales down relative logit differences in the exponent, spreading probability mass more evenly across options without altering which logit is highest.

D. The logit scores are recalculated, changing which token holds the highest rank in the vocabulary.

Temperature applies an ideal transformation to fixed logits during softmax calculation; it does not retrain the model or reorder logit rankings.

Question 6

Correct answer: D

A. The tokenizer vocabulary is updated to permanently add new vocabulary IDs found in the document.

Tokenizers possess fixed vocabularies established prior to model deployment; RAG does not rewrite tokenizer vocabulary tables.

B. The system uses the retrieved passage to execute a fast fine-tuning step that updates the model's weights.

Retrieving documents alters the contextual input for an inference step but does not update or fine-tune neural network parameters.

C. The model's causal attention mask is disabled to allow direct insertion of missing factual information into output logits.

Causal attention masking remains active during generation to enforce step-by-step autoregressive context constraints.

D. Selected external text passages are inserted into the prompt context for the fixed model to use during inference.

RAG retrieves relevant external text and appends it to the context window supplied to the model, leaving parameter weights untouched.

Source and derivation

The source is the original English-language NotebookLM Quiz created from the selected Foundations of Generative Artificial Intelligence book. The native question interface, hints and answer feedback were observed directly. No questions, options, rationales or numeric values were added or recalculated for this PDF.

Original native quiz artifact (the source interface may require its institutional account). This PDF can be read independently.